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Studierichting: Informatica
Oefeningenreeks 4A nr. 43.3

$$I = \int \frac{4x^2 - 6x + 1}{2x^3 - x^2}\,dx$$
$$\frac{4x^2 - 6x + 1}{x^2(2x-1)} = \frac{A}{x^2} + \frac{B}{x} + \frac{C}{2x-1}$$
$$A = \frac{4x^2 - 6x + 1}{2x-1}|_{x=0} = \frac{1}{-1} = -1$$
$$\frac{4(x-1)}{x(2x-1)} = \frac{B}{x} + \frac{C}{2x-1}$$
$$B = \frac{4(x-1)}{2x-1}|_{x=0} = 4$$
$$C = \frac{4(x-1)}{x}|_{x=\frac{1}{2}} = -4$$
$$I = \int \left(-\frac{1}{x^2} + \frac{4}{x} - \frac{4}{2x-1}\right) dx$$
$$I = \frac{1}{x} + 4\ln|x| - 2\ln|2x-1| + C$$

References